

CHAPTER 4

INDUCTION AND RECURRENCES

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4.1 Induction

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4.1.1 Ordinary induction

The Induction Principle.

Let P be a predicate on nonnegative integers. If

- $P(0)$ is true, and
- $P(n)$ IMPLIES $P(n + 1)$ for all nonnegative integers n

then

- $P(m)$ is true for all nonnegative integers m .

■ Example: Prove that for all $n \in \mathbb{N}$, we have

$$a) 1 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$b) 1 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

4.1.2 Strong Mathematical Induction

Theorem. Let $P(n)$ be a predicate with the universe of discourse $\mathbf{N}$. Assume that

- $P(0)$ is true
- $\forall n, (P(0) \wedge P(1) \wedge \dots \wedge P(n)) \rightarrow P(n+1)$

Then $\forall n, P(n)$ is true

In our proofs, to show $P(k+1)$, our inductive hypothesis assures that ALL of $P(0), P(1), \dots P(k)$ are true, so we can use ANY of them to make the inference.

■ Prove that the Fibonacci sequence

$$a_0 = 1, a_1 = 1,$$

$$a_{k+1} = a_k + a_{k-1} \text{ for } k \geq 1$$

satisfies

$$a_k \geq \left(\frac{3}{2}\right)^{k-2} \quad \forall k \geq 1.$$

Problem 5.29.

The following Lemma is true, but the *proof* given for it below is defective. Pinpoint *exactly* where the proof first makes an unjustified step and explain why it is unjustified.

Lemma. *For any prime p and positive integers $n, x_1, x_2, \dots, x_n$, if $p \mid x_1 x_2 \dots x_n$, then $p \mid x_i$ for some $1 \leq i \leq n$.*

Bogus proof. Proof by strong induction on n . The induction hypothesis $P(n)$ is that Lemma holds for n .

Base case $n = 1$: When $n = 1$, we have $p \mid x_1$, therefore we can let $i = 1$ and conclude $p \mid x_i$.

Induction step: Now assuming the claim holds for all $k \leq n$, we must prove it for $n + 1$.

So suppose $p \mid x_1 x_2 \dots x_{n+1}$. Let $y_n = x_n x_{n+1}$, so $x_1 x_2 \dots x_{n+1} = x_1 x_2 \dots x_{n-1} y_n$. Since the right-hand side of this equality is a product of n terms, we have by induction that p divides one of them. If $p \mid x_i$ for some $i < n$, then we have the desired i . Otherwise $p \mid y_n$. But since y_n is a product of the two terms x_n, x_{n+1} , we have by strong induction that p divides one of them. So in this case $p \mid x_i$ for $i = n$ or $i = n + 1$. ■

4.1.3 Inductive Definitions

We completely understand the function $f(n) = n!$, right?

As a reminder, here's the definition:

$$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-1) \cdot n, n \geq 1$$

But equivalently, we could define it like this:

$$n! = \begin{cases} n \cdot (n-1)!, & \text{if } n > 1 \\ 1, & \text{if } n = 1 \end{cases}$$

Recursive Case

Base Case

Inductive (Recursive) Definition

4.1.3 Inductive Definitions

Another VERY common example:

Fibonacci Numbers

$$F(n) = \begin{cases} 0 & \text{if } n = 0 \\ 1 & \text{if } n = 1 \\ F(n-1) + F(n-2) & \text{if } n > 1 \end{cases}$$

Base Cases

Recursive Case

Is there a non-recursive definition for the Fibonacci Numbers?

Fibonacci numbers

$$F(0) ::= 0,$$

$$F(1) ::= 1,$$

$$F(n) ::= F(n-1) + F(n-2) \quad \text{for } n \geq 2.$$

Exercise: Prove, using strong induction, the following closed-form formula for the Fibonacci numbers

$$F(n) = \frac{p^n - q^n}{\sqrt{5}}$$

$$\text{where } p = \frac{1+\sqrt{5}}{2} \text{ and } q = \frac{1-\sqrt{5}}{2}.$$

Hint: Note that p and q are the roots of $x^2 - x - 1 = 0$, and so $p^2 = p + 1$ and $q^2 = q + 1$.

H int

$$\text{Prove: } F(0) = \frac{p^0 - q^0}{\sqrt{5}}$$

$$p^0 = 1 \text{ and } q^0 = 1 \text{ so } \frac{p^0 - q^0}{\sqrt{5}} = 0 = F(0)$$

$$\text{Prove: } F(1) = \frac{p^1 - q^1}{\sqrt{5}}$$

...

$$\text{Assume } F(n) = \frac{p^n - q^n}{\sqrt{5}} \text{ for } 2 \leq n < k. \text{ Prove } F(k) = \frac{p^k - q^k}{\sqrt{5}}$$

$$F(k) = F(k-1) + F(k-2) = \frac{p^{k-1} - q^{k-1}}{\sqrt{5}} + \frac{p^{k-2} - q^{k-2}}{\sqrt{5}} = \frac{p^{k-2}(p+1) + q^{k-2}(q+1)}{\sqrt{5}} = \frac{p^k - q^k}{\sqrt{5}}$$

4.2 Recurrences

- A recurrence describes a sequence of numbers. Early terms are specified explicitly, and later terms are expressed as a function of their predecessors.
- Example: Fibonacci numbers

$$F(0) ::= 0,$$

$$F(1) ::= 1,$$

$$F(n) ::= F(n-1) + F(n-2) \quad \text{for } n \geq 2.$$

Example. How many bit strings of length n do not contain two consecutive bit 0?

■ Let a_n be the number of such strings, and $x_1x_2 \dots x_n$ be such a string:

➤ if $x_n = 1$ then the string is $x_1x_2 \dots x_{n-1}1$, where $x_1x_2 \dots x_{n-1}$ is an arbitrary string of length $n-1$ without two consecutive bit 0: there are a_{n-1} such strings

➤ if $x_n = 0$ then the string is $x_1x_2 \dots x_{n-2}10$, where $x_1x_2 \dots x_{n-2}$ is an arbitrary string of length $n-2$ without two consecutive bit 0: there are a_{n-2} such strings

Thus
$$a_n = a_{n-1} + a_{n-2}$$

■ This is an example of a **recurrence relation**

Modeling with Recurrence Relations

Example: A sum of P_0 is deposited in the saving account with the interest rate of r *compounded* annually. How much will be in the account after n years?

Solution:

Let P_n be the amount after n years, then P_n is obtained from P_{n-1} and the interest yielded from that amount in one year:

$$P_n = P_{n-1} + r P_{n-1} = (1+r) P_{n-1}$$

Modeling with Recurrence Relations

Solution:

Let P_n be the amount after n years, then :

$$P_n = P_{n-1} + r P_{n-1} = (1+r) P_{n-1}$$

This is a *geometric progression*.

The solution of this relation is obtained as follows:

$$P_n = (1+r) P_{n-1} = (1+r)^2 P_{n-2} = \dots = (1+r)^n P_0$$

$$P_n = (1+r)^n P_0$$

Modeling with Recurrence Relations

- In this section, we'll emphasize using recurrences to analyze the performance of recursive algorithms.
- Recurrences are one aspect of a broad theme in computer science: reducing a big problem to progressively smaller problems until easy base cases are reached. This same idea underlies both induction proofs and recursive algorithms. As we'll see, all three ideas snap together nicely. For example, the running time of a recursive algorithm could be described with a recurrence with induction used to verify the solution.

- There are several methods for solving recurrence equations.
- The simplest is to guess the solution and then verify that the guess is correct with an induction proof.

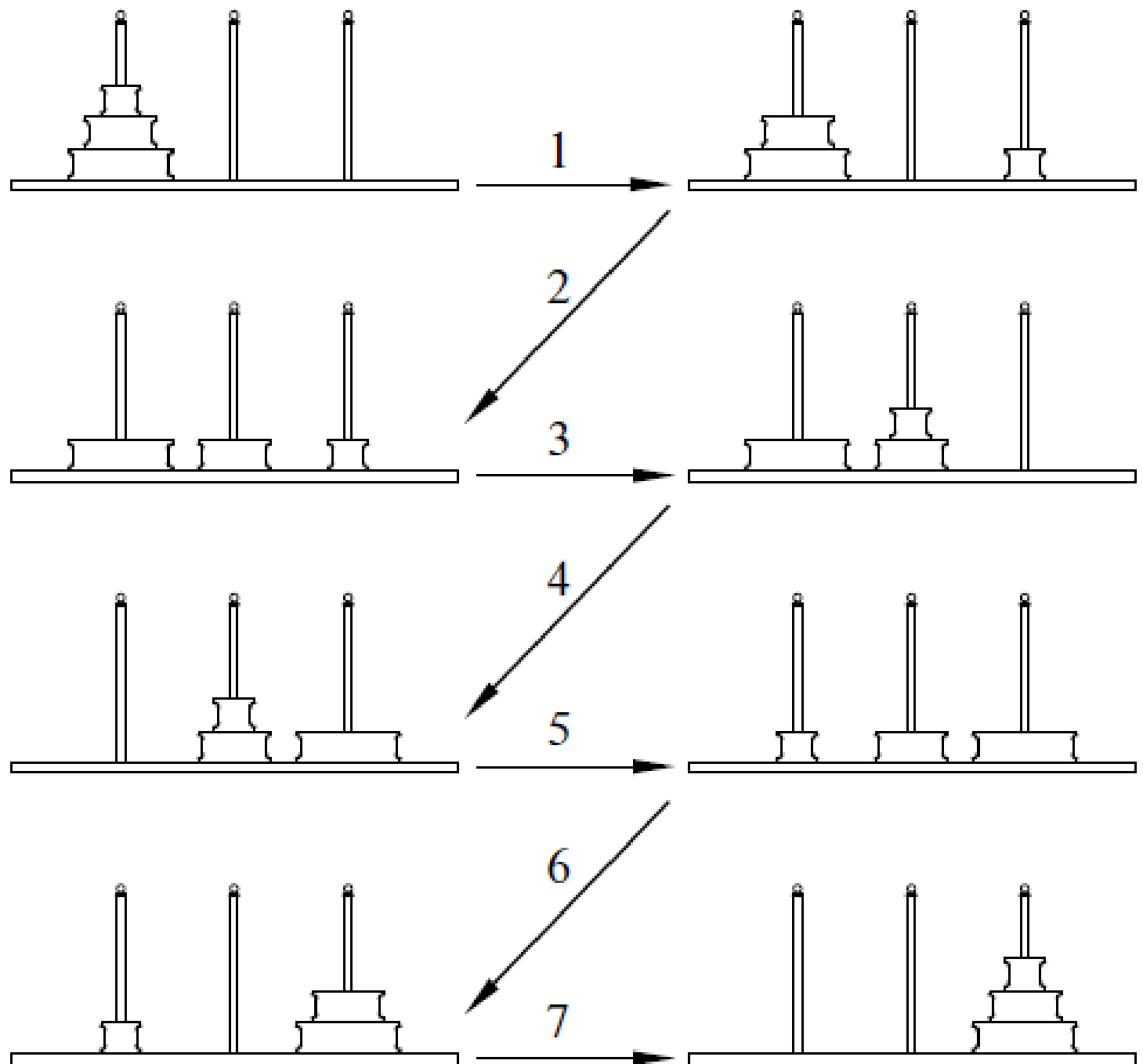
The Towers of Hanoi Problem

- According to legend, there is a temple in Hanoi with three posts and 64 gold disks of different sizes. Each disk has a hole through the center so that it fits on a post.
- In the misty past, all the disks were on the first post, with the largest on the bottom and the smallest on top.

- Monks in the temple have labored through the years since to move all the disks to one of the other two posts according to the following rules:
 - The only permitted action is removing the top disk from one post and dropping it onto another post.
 - A larger disk can never lie above a smaller disk on any post.
- The legend says that when the monks complete the puzzle, the world will end!

Question

- “Given sufficient time, can the monks succeed?”
- If so, “How long until the world ends?”



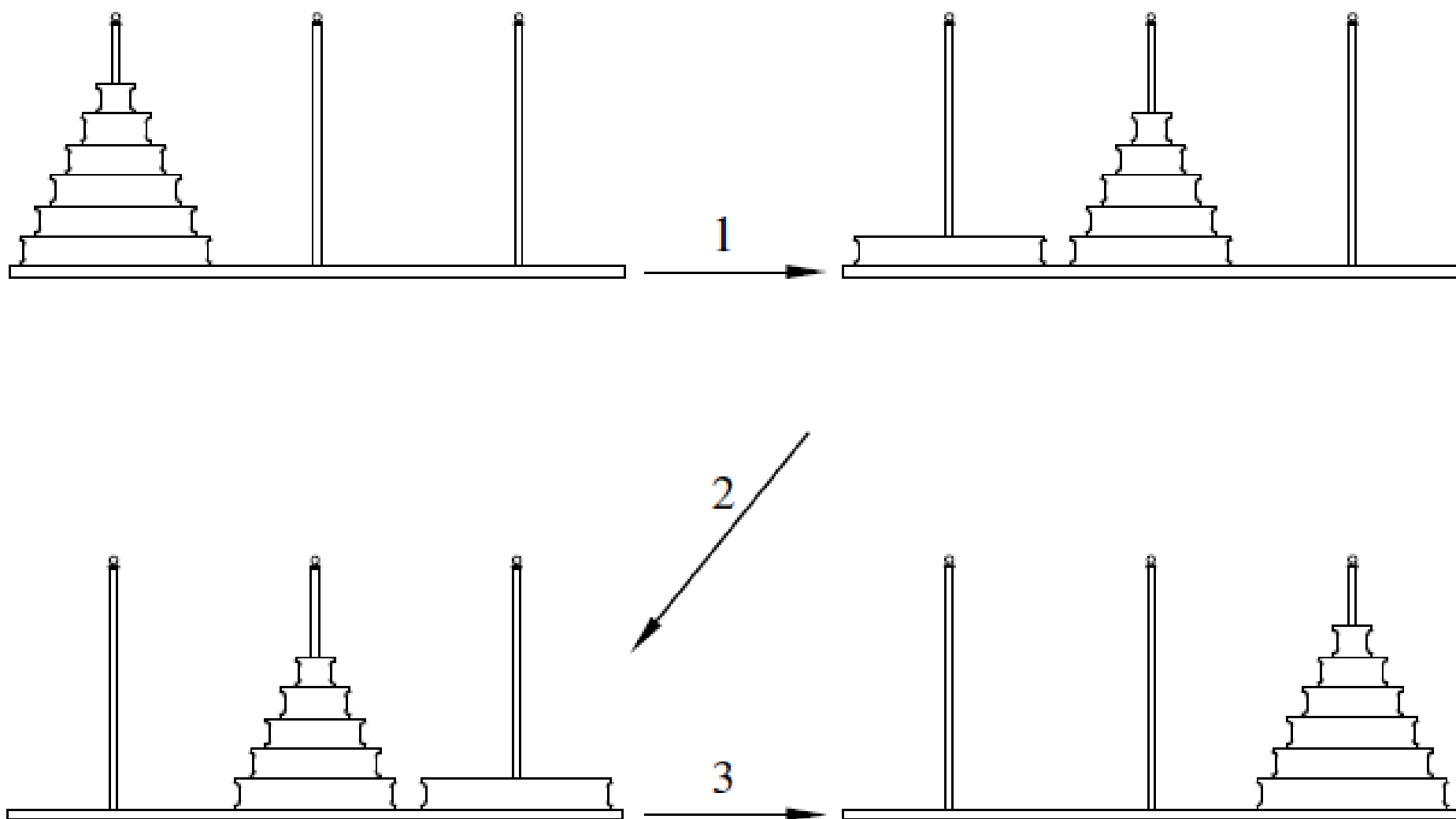


Figure 16.3 A recursive solution to the Towers of Hanoi problem.

A recursive solution

- **Stage 1.** Move the top $n-1$ disks from the first post to the second using the solution for $n-1$ disks. This can be done in t_{n-1} steps.
- **Stage 2.** Move the largest disk from the first post to the third post. This takes just 1 step.
- **Stage 3.** Move the $n-1$ disks from the second post to the third post, again using the solution for $n-1$ disks. This can also be done in t_{n-1} steps.
- This algorithm shows that t_n , the minimum number of steps required to move n disks to a different post, is at most $t_{n-1} + 1 + t_{n-1} = 2t_{n-1} + 1$.

Find a closed formula for T_n

- We have $T_n = 2 T_{n-1} + 1$
- Let's guess the formula

$$T_1 = 1$$

$$T_2 = 3$$

$$T_3 = 7$$

$$T_4 = 15$$

$$T_5 = 31$$

Claim 22.1.1. $T_n = 2^n - 1$ satisfies the recurrence:

$$T_1 = 1$$

$$T_n = 2T_{n-1} + 1 \quad (\text{for } n \geq 2).$$

Proof. The proof is by induction on n . The induction hypothesis is that $T_n = 2^n - 1$. This is true for $n = 1$ because $T_1 = 1 = 2^1 - 1$. Now assume that $T_{n-1} = 2^{n-1} - 1$ in order to prove that $T_n = 2^n - 1$, where $n \geq 2$:

$$\begin{aligned} T_n &= 2T_{n-1} + 1 \\ &= 2(2^{n-1} - 1) + 1 \\ &= 2^n - 1. \end{aligned}$$

The first equality is the recurrence equation, the second follows from the induction assumption, and the last step is simplification. ■

MERGE SORT ALGORITHM

```
procedure mergesort( $L = a_1, \dots, a_n$ )  
  if  $n > 1$  then  
     $m := \lfloor n/2 \rfloor$   
     $L_1 := a_1, a_2, \dots, a_m$   
     $L_2 := a_{m+1}, a_{m+2}, \dots, a_n$   
     $L := \text{merge}(\text{mergesort}(L_1), \text{mergesort}(L_2))$   
  {  $L$  is now sorted into elements in nondecreasing order }
```

The input is a list of n numbers, and the output is those same numbers in nondecreasing order.

Merging two lists

procedure *merge*($L1, L2$: sorted lists)

■ $L :=$ empty list

■ **while** $L1$ and $L2$ are both nonempty

Remove the smaller of first elements of $L1$ and $L2$ from its list; put it at the right end of L

*If this removal makes one list empty **then** remove all elements from the other list and append them to L*

■ **return** L

(L is the merged list with elements in increasing order)

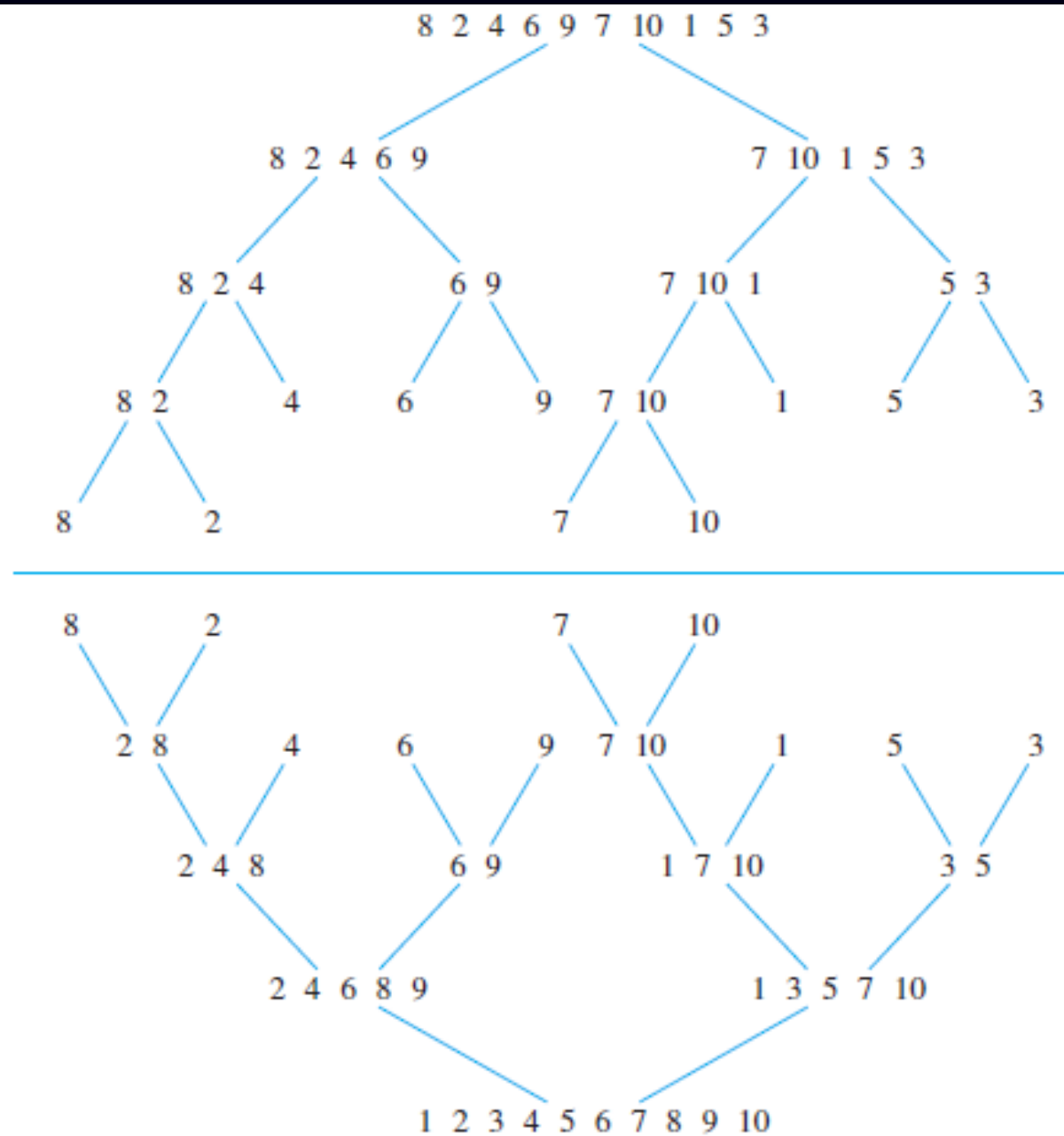


FIGURE 2 The Merge Sort of 8, 2, 4, 6, 9, 7, 10, 1, 5, 3.

Merge sort

- What is the maximum number of comparisons used in sorting n items?

Finding a recurrence

Let T_n be the maximum number of comparisons used while Merge Sorting a list of n numbers.

Assume that n is a power of 2 (so the input can be divided in half at every stage of the recursion).

- If there is only one number in the list, then no comparisons are required, so $T_1 = 0$.
- Otherwise, T_n includes comparisons used in sorting the first half (at most $T_{n/2}$), in sorting the second half (also at most $T_{n/2}$), and in merging the two halves. The number of comparisons in the merging step is at most $n-1$.

- For $n \geq 2$ and a power of 2

$$T_1 = 0$$

$$T_n = 2T_{n/2} + n - 1$$

- The closed-form for T_n

$$T_n = n \log n - n + 1$$

Solving linear recurrence

Definition: A *linear homogeneous recurrence relation of degree k* is a recurrence relation of the form

$$f(n) = c_1 f(n-1) + c_2 f(n-2) + \cdots + c_k f(n-k)$$

where $c_1, c_2, \dots, c_k$ are real numbers, $c_k \neq 0$.

Example: Fibonacci

Solving linear recurrence

Find the explicit formula for $f(n)$

$$f(n) = c_1 f(n-1) + c_2 f(n-2)$$

- Solve the characteristic equation

$$r^2 - c_1 r - c_2 = 0$$

- Case 1: The characteristic equation has two distinct root r_1, r_2 .

$$f(n) = \alpha_1 r_1^n + \alpha_2 r_2^n \quad (\alpha_1, \alpha_2 \in \mathbb{R})$$

- Case 2: The characteristic equation has only one root r_0 .

$$f(n) = \alpha_1 r_0^n + \alpha_2 n r_0^n \quad (\alpha_1, \alpha_2 \in \mathbb{R})$$

Example

What is the solution of the recurrence relation

$$a_n = a_{n-1} + 2a_{n-2}$$

with $a_0 = 2$ and $a_1 = 7$?

Solution.

The characteristic equation : $r^2 - r - 2 = 0$ has distinct roots $r_1 = 2$ and $r_2 = -1$. Therefore,

$$a_n = c_1 2^n + c_2 (-1)^n$$

for some c_1, c_2 . From the initial conditions, it follows that

$$a_0 = 2 = c_1 2^0 + c_2 (-1)^0$$

$$a_1 = 7 = c_1 2 + c_2 (-1)$$

So we get $c_1 = 3, c_2 = -1$. Hence, the solution of the recurrence relation is

$$a_n = 3 \cdot 2^n - (-1)^n.$$

- Find the closed formula for Fibonacci numbers

$$F(0) = 0$$

$$F(1) = 1$$

$$F(n) = F(n-1) + F(n-2)$$

Example

What is the solution of the recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2}$$

with initial conditions $a_0 = 1$ and $a_1 = 6$?

Solving general linear recurrence

Theorem. Let $c_1, c_2, \dots, c_k$ be real numbers.

Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has k distinct roots $r_1, \dots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$$

for $n = 0, 1, 2, \dots$, where $\alpha_1, \alpha_2, \dots, \alpha_k$ are constants.

Example

Find the solution to the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$$

with the initial conditions $a_0 = 2$, $a_1 = 5$, and $a_2 = 15$.

Theorem

Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has t distinct roots $r_1, r_2, \dots, r_t$ with multiplicities $m_1, m_2, \dots, m_t$, respectively, so that $m_i \geq 1$ for $i = 1, 2, \dots, t$ and $m_1 + m_2 + \dots + m_t = k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$$\begin{aligned} a_n = & (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n \\ & + \dots + (\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n \end{aligned}$$

for $n = 0, 1, 2, \dots$, where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_i - 1$.

Example

Find the solution to the recurrence relation

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$$

with initial conditions $a_0 = 1$, $a_1 = -2$, and $a_2 = -1$.

Theorem

Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has t distinct roots $r_1, r_2, \dots, r_t$ with multiplicities $m_1, m_2, \dots, m_t$, respectively, so that $m_i \geq 1$ for $i = 1, 2, \dots, t$ and $m_1 + m_2 + \dots + m_t = k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if

$$\begin{aligned} a_n = & (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n \\ & + \dots + (\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n \end{aligned}$$

for $n = 0, 1, 2, \dots$, where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_i - 1$.

Solving inhomogeneous linear recurrence

$$f(n) = a_1 f(n-1) + a_2 f(n-2) + \cdots + a_d f(n-d) + g(n).$$

1. Find roots of the characteristic equation.
2. Write down the homogeneous solution to the homogeneous recurrence

$$F(n) = a_1 F(n-1) + a_2 F(n-2) + \cdots + a_d F(n-d)$$

3. Now restore $g(n)$ and find a single solution $G(n)$ to the recurrence, ignoring boundary conditions. This is called a particular solution.
4. Add the homogeneous and particular solutions together to obtain the general solution $F(n) + G(n)$
5. Now use the boundary conditions to determine constants.

■ Example: Solve the recurrence relation

$$f(1) = 1$$

$$f(n) = 2f(n-1) + n \quad \text{for } n \geq 2.$$

■ Solution

➤ Solving the homogeneous recurrence

$$F(n) = 2F(n-1)$$

we get the homogeneous solution: $F(n) = c \cdot 2^n$

➤ Let's guess that there is a particular solution of the form $G(n) = a \cdot n + b$

Substituting this guess into the recurrence gives

$$an + b = 2(a(n-1) + b) + n$$

$$0 = (a+1)n + (b-2a).$$

We deduce that $a = -1$, $b = -2$

- The general solution to the recurrence

$$f(n) = c2^n - n - 2.$$

- Using the boundary condition $f(1) = 1$, we get

$$1 = f(1) = c \cdot 2 - 1 - 2$$

So $c = 2$.

Therefore, $f(n) = 2 \cdot 2^n - n - 2$.

Solution of well-known recurrences

	Recurrence	Solution
Towers of Hanoi	$T_n = 2T_{n-1} + 1$	$T_n \sim 2^n$
Merge Sort	$T_n = 2T_{n/2} + n - 1$	$T_n \sim n \log n$
Hanoi variation	$T_n = 2T_{n-1} + n$	$T_n \sim 2 \cdot 2^n$
Fibonacci	$T_n = T_{n-1} + T_{n-2}$	$T_n \sim (1.618\dots)^{n+1} / \sqrt{5}$

Solve these recurrence relations together with the initial conditions given.

a) $a_n = 2a_{n-1}$ for $n \geq 1$, $a_0 = 3$

b) $a_n = a_{n-1}$ for $n \geq 1$, $a_0 = 2$

c) $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$, $a_0 = 1$, $a_1 = 0$

d) $a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2$, $a_0 = 6$, $a_1 = 8$

e) $a_n = -4a_{n-1} - 4a_{n-2}$ for $n \geq 2$, $a_0 = 0$, $a_1 = 1$

Solve these recurrence relations together with the initial conditions given.

a) $a_n = a_{n-1} + 6a_{n-2}$ for $n \geq 2$, $a_0 = 3$, $a_1 = 6$

b) $a_n = 7a_{n-1} - 10a_{n-2}$ for $n \geq 2$, $a_0 = 2$, $a_1 = 1$

c) $a_n = 6a_{n-1} - 8a_{n-2}$ for $n \geq 2$, $a_0 = 4$, $a_1 = 10$

d) $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 2$, $a_0 = 4$, $a_1 = 1$

e) $a_n = a_{n-2}$ for $n \geq 2$, $a_0 = 5$, $a_1 = -1$

f) $a_n = -6a_{n-1} - 9a_{n-2}$ for $n \geq 2$, $a_0 = 3$, $a_1 = -3$

g) $a_{n+2} = -4a_{n+1} + 5a_n$ for $n \geq 0$, $a_0 = 2$, $a_1 = 8$